Dräger

Dräger Evita® V600
ICU Ventilation and Respiratory Monitoring

Experience the next level of ventilator operation. The Evita® V600 combines high performance ventilation with an aesthetic design enabling quick and efficient operation. From the first onset of a lung protective ventilation until the integration of a patient care-centred intensive care workplace.

D-5742-2018

Benefits

Operation principle and user interface

The brilliant user interface combined with up-to-date glass touch technology supports intuitive operation.

- Quick and safe to operate even in the most stressful situations due to intuitive menu access to both settings and your clinical data.
- All patient data, alarms and trends are fully recorded. Conveniently exported via USB interface.
- Switch between multiple view configurations with the touch of a finger.
- Step-by-step guidance leads you through every procedure.
- Easy to read and navigate thanks to our new colour concept and glass touch display.
- The 360° alarm light flashes in the colour of the corresponding alarm priority and is visible from every direction.

Early mobilization and patient transport

Support of instant, flexible patient transport through optional hardware components and early mobilization.

- Gas supply unit GS500
- Power supply unit PS500
- Bed Coupling
- Transport supply unit TSU

Non-invasive, for as long as possible

Safeguarding High Flow Therapy

- Support of high flow nasal cannula (HFNC) up to 80 l/min with possibility to limit the maximum pressure
- Provision of high flow oxygen even with active humidification
- Smooth and seamless transition from O_2 therapy through NIV to invasive ventilation and back

NIV with automatic leak compensation

- Non-invasive ventilation (NIV) in all modes
- Automatic leak compensation complements the patient's flow demand with continuous breathing gas
- Adaptation for leakages related to trigger, cycling, application of pressures and volumes
- Not applicable alarms not displayed during NIV
- Correct monitoring of patient ventilation due to leak compensated values

Lung protective ventilation

Our comprehensive treatment tools help support your lung protective ventilation strategy.

- Lung protective ventilation for adults, children and neonates: invasive, non-invasive and with O_2-therapy

Benefits

- Advanced lung monitoring and diagnostic features (e.g. Smart Pulmonary View or Low Flow manoeuvre)
- Display of regional ventilation distribution with PulmoVista® 500
- Recruitment tools (e.g. QuickSet® and PressureLink) and therapy decision support with breath-by-breath trends (PEEP, EIP, Vt, C_{dyn})
- Focus on end-expiratory lung volume with PC-APRV with AutoRelease®
- Volumetric CO_2-Monitoring (VCO_2, $VTCO_2$, Slope Phase 3, Vds/VTe)

Effective weaning

Support of a synchronized way for quick and efficient weaning.

- Automatic weaning with SmartCare®/PS
- Increased variability in spontaneous breathing through Variable Pressure Support or Proportional Pressure Support
- Facilitating spontaneous breathing through “room to breathe concept” with e.g. AutoFlow® or Volume Guarantee
- Automatic tube compensation (ATC®) compensates for artificial airway resistance
- Assessment of weaning through RSBi, P0.1 and NIF

Medical device interoperability and cybersecurity

We envision a future of acute care where medical devices are connected as a system and allow to enable new clinical applications in a safe and secure environment.
Interoperability between medical devices supports caregivers at the point of care:

- Future-proof open connectivity thanks to standardised communication based on ISO/IEEE 11073-SDC-standard principles*
- Automatic documentation from medical devices to your electronic medical records (EMR)**
- Alarm distribution: Support patient safety by notifying caregivers with a Distributed Information System.
- Central monitoring of ventilation parameters with the VentCentral application at the Infinity® CentralStation

* only with Connectivity Converter CC300

** only with Connectivity Converter CC300 and Infinity® Gateway Suite

We strive to consistently implement measures according to the NIST security best practices framework. Following the five functions, this encompasses:

Identify
Dedicated documents with security relevant information for your asset risk management (e.g. Software Bill Of Material, MDS2 Form, comprehensive cybersecurity whitepaper).

Protect

- Secure Boot ensures the integrity of the software running on the device

Benefits

- Access control for protected functions and data
- Hardened operating system by omitting unnecessary software components and disabling all unused ports to minimize attack surface

Detect
Security relevant events are detected, logged in a tamper proof security log file and the IT-admin is notified via SNMP traps

Respond
The system health monitor observes the system load carefully and reacts in case of suspected malicious events, i.e. by disabling the network interface if load is unusually high.

Recover
System can reboot into last good known state if a security event is detected. Dräger service can restore hard- and software quickly, clinical configuration can be transferred from other devices via USB drive.

Dräger Services that go beyond repairing equipment

Your medical equipment performs at its best when correctly calibrated and regularly maintained by original manufacturer service:

- TotalCare: Budget security on maintenance and repair.
- PreventiveCare: Avoid unexpected failures in advance.
- InspectionCare: Regular inspections to ensure the secure operation of your devices.
- ExtendedCare: Coverage beyond the standard warranty period.

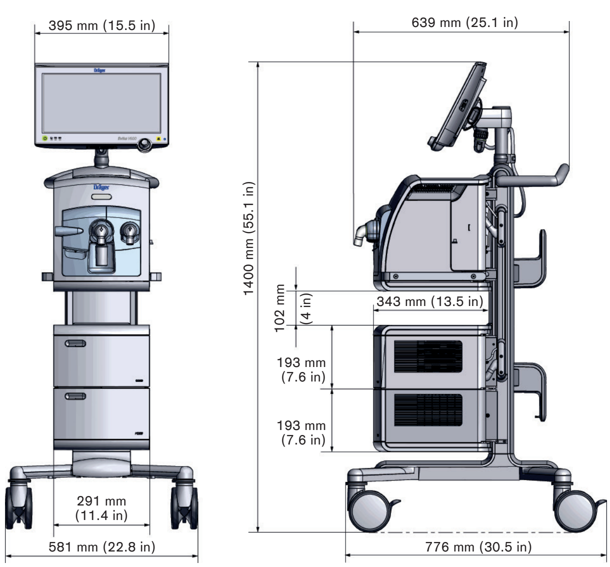
Increase the availability of both your medical equipment and IT solutions – keeping them updated, safe and secure with our Connected Maintenance offerings:

- Help Ticket
- Software Distribution
- Certificate Management

Awards

iF DESIGN AWARD 2020

Physical Specifications



Designation	Weight
Ventilation unit	approx. 16kg (34.9 lbs)
Display unit	approx. 5 kg (11 lbs)
Ventilation unit and display unit	approx. 21 kg (46.3 lbs)
Trolley	approx. 33 kg (72.8 lbs)
PS500	approx. 16 kg (35.2 lbs)
GS500	approx. 10.5 kg (23 lbs)
Nominal weight (weight of the ventilation unit and display unit on trolley)	58 kg (128 lbs)
Maximum weight (permitted maximum total weight)	133 kg (293 lbs)

Dimensions and weights of the Evita V600

Accessories

D-3491-2019

Ventilation accessories

Day in, day out in the hospital you are faced with enormous time and cost pressures at the same time you are taking care of the welfare of your patients. You need technical medical accessories which you can use to unlock the full potential of your equipment, which work smoothly, ensure the best possible care of your patients and help you to improve your processes. In short: accessories you can rely on, Dräger can supply it for you. Find them in our accessory catalogue.

Related Products

D-23101-2020

Dräger PulmoVista® 500

Making ventilation visible. Put the power of Electrical Impedance Tomography (EIT) to work for you and your patients. With the PulmoVista® 500, you can visualise regional ventilation distribution within the lungs – non-invasive, in real time and directly at bedside.

D-12504-2014

Nitric oxide therapy

The NO application device NO-A is a therapy device for the application of NO therapy gas in the ventilation flow of an intensive care patient in connection with several Dräger ventilation units. NO inhalation therapy reduces the clinical necessity of a right-to-left shunt and contributes to cardiac relief. In order to improve the oxygenation of ventilated patients nitrogen monoxide (NO) is added to the respiratory gas. Inhaled NO has a dilating effect on the blood vessels in the lung. The target concentration is set on NO-A and the delivered gas volume is automatically and electronically adjusted to the values of the ventilation system via Medibus connection.

Related Products

D-6036-2018

Connectivity Converter CC300

Enable seamless and secure data export from a medical device with the Connectivity Converter CC300. This system module allows standardised information to be shared amongst medical devices and with hospital information systems. The electronic documentation functionality increases efficiency of administrative workflows at the point-of-care.

D-30093-2020

Dräger Alarm and Information System

Your solution to distribute alerts to assigned caregivers in the unit. The Distributed Information System (DIS) provides additional alarm information for clinical staff in the patient vicinity. Caregivers are location-independently informed about point-of-care alarms from Dräger patient ventilators and monitors via central stations and mobile phones.

Technical Data

Patient type	Adults, paediatric patients, neonates
Ventilation settings	
Ventilation mode	Volume controlled ventilation: – VC-CMV – VC-SIMV – VC-AC – VC-MMV
	Pressure controlled ventilation: – PC-CMV – PC-BIPAP[1] / SIMV+ – PC-SIMV – PC-AC – PC-APRV – PC-PSV
	Support of spontaneous breathing: – SPN-CPAP/PS – SPN-CPAP/VS – SPN-CPAP – SPN-PPS
Enhancements	– AutoFlow® [1]/ Volume Guarantee – Variable Pressure Support – Smart Pulmonary View – Automatic Tube Compensation (ATC® [1]) – SmartCare®/PS 2.0 – Automated clinical protocol in SPN-CPAP/PS – Low Flow PV Loop
Special procedures	– Suction manoeuvre – Manual inspiration/hold – Medication nebulisation – P0.1 – PEEPi – NIF
Therapy types	– Invasive ventilation (Tube) – Non-invasive ventilation (NIV) – O_2-therapy
Respiratory rate (RR)	Adult 0.5 to 98/min Paediatric patients, Neonates 0.5 to 150/min
Inspiratory time (Ti)	Adults 0.11 to 10 s Paediatric patients, Neonates 0.1 to 10 s
Tidal volume (VT)	Adults 0.1 to 3.0 L Paediatric patients 0.02 to 0.3 L Neonates 0.002 to 0.1 L
Inspiratory flow (Flow)	Adults 2 to 120 L/min Paediatric patients 2 to 30 L/min
Maximum flow during non-invasive ventilation of neonates (Flow max)	0 to 30 L/min
Inspiratory pressure (Pinsp)	1 to 95 mbar (or hPa or cmH_2O)
Pressure limitation (Pmax)	2 to 100 mbar (or hPa or cmH_2O)
Positive end-expiratory pressure (PEEP)	0 to 50 mbar (or hPa or cmH_2O)
Additional intermittent PEEP for sighs (ΔintPEEP)	0 to 20 mbar (or hPa or cmH_2O)

Technical Data

Pressure support (Psupp)	0 to 95 mbar (or hPa or cmH_2O)
Pressure rise time (Slope)	Adults, Paediatric patients 0 to 2 s Neonates 0 to 1.5 s
O_2 concentration (FiO_2)	21 to 100 Vol. %
Trigger threshold (Flow trigger)	0.2 to 15 L/min
Automatic Tube Compensation (ATC®)	Inside tube diameter tube Ø – Endotracheal tube ET Adults 5 to 12 mm (0.2 to 0.47 inch) Paediatric patients 2 to 8 mm (0.08 to 0.31 inch) Neonates 2 to 5 mm (0.08 to 0.2 inch) – Tracheostoma tube (Trach.) Adults 5 to 12 mm (0.2 to 0.47 inch) Paediatric patients 2.5 to 8 mm (0.1 to 0.31 inch) – Degree of compensation 0 to 100 %

Airway Pressure Release Ventilation (APRV)	
Inspiratory time (Thigh)	0.1 to 30 s
Expiratory time (Tlow)	0.05 to 30 s
Maximum duration of lower pressure level (Tlow max)	0.05 to 30 s
Upper pressure level (Phigh)	1 to 95 mbar (or hPa or cmH_2O)
Lower pressure level (Plow)	0 to 50 mbar (or hPa or cmH_2O)
Expiration termination criterion (in relation to the peak expiratory flow) (Exp. term.)	1 to 80 % (PEF)

Proportional Pressure Support (SPN-PPS)	
Flow-based assistance (Flow Assist)	Adults 0 to 30 mbar/L/s (or hPa/L/s or cmH_2O/L/s) Paediatric patients 0 to 100 mbar/L/s (or hPa/L/s or cmH_2O/L/s) Neonates 0 to 300 mbar/L/s (or hPa/L/s or cmH_2O/L/s)
Volume-based assistance (Vol. Assist)	Adults 0 to 100 mbar/L (or hPa/L or cmH_2O/L)
corresponds to compliance compensation	10000 to 10 mL/mbar (or mL/hPa or mL/cmH_2O) Paediatric patients 0 to 1000 mbar/L (or hPa/L or cmH_2O/L)
corresponds to compliance compensation	10000 to 1 mL/mbar (or mL/hPa or mL/cmH_2O) Neonates 0 to 4000 mbar/L (or hPa/L or cmH_2O/L)
corresponds to compliance compensation	1000 to 0.3 mL/mbar (or mL/hPa or mL/cmH_2O)

O_2-therapy	Continuous Flow 2-80* L/min (adults), 2-30 L/min (paediatric patients), 2-15 L/min (neonates) * depending on local approval, otherwise 60 l/min Pmax 2-55 mbar O_2 concentration, FiO_2 21 to 100 Vol%
Leakage compensation	On/Off On: full compensation active Off: only trigger compensation active

Technical Data

Displayed measured values	
Airway pressure measurement	Plateau pressure (Pplat) Positive end-expiratory pressure (PEEP) Peak Inspiratory Pressure (PIP) Mean airway pressure (Pmean) Minimum airway pressure (Pmin) Range -60 to 120 mbar (or hPa or cmH_2O)
Flow measurement	
Minute volume measurement	Expiratory minute volume, overall, not leakage-corrected (MVe) Inspiratory minute volume, overall, not leakage-corrected (MVi) Minute volume, leakage-corrected (MV) Mandatory expiratory minute volume, overall, not leakage-corrected (MVemand) Spontaneous expiratory minute volume, overall, not leakage-corrected (MVespon) Range 0 to 99 L/min BTPS
Tidal volume measurement	Tidal Volume, leakage-corrected (VT) Mandatory inspiratory tidal volume, not leakage-corrected (VTimand) Mandatory expiratory tidal volume, not leakage-corrected (VTemand) Spontaneous inspiratory tidal volume, not leakage-corrected (VTispon) Range 0 to 5500 mL BTPS
Respiratory rate measurement	Respiratory rate (RR) Mandatory respiratory rate (RRmand) Spontaneous respiratory rate (RRspon) Range 0/min to 300/min
O_2 measurement (inspiratory side)	Inspiratory O_2 concentration (in dry air) (FiO_2) Range 18 to 100 Vol %
CO_2 measurement in main flow (adults and paediatric patients only)	End-tidal CO_2 concentration ($etCO_2$) Range 0 to 120 mmHg

Displayed calculated values	
Dynamic compliance (Cdyn)	Range 0 to 650 mL/mbar (or mL/hPa or mL/cmH_2O)
Resistance (R)	Range 0 to 1000 mbar/L/s (or hPa/L/s or cmH_2O/L/s)
Leakage minute volume (MVleak)	Range 0 to 99 L/min, BTPS
Rapid shallow breathing index (RSBI)	Adults 0 to 9999 (/min/L) Paediatric patients 0 to 9999 (/min/L) Neonates 0 to 300 (/min/L)
Negative Inspiratory Force (NIF)	Range -80 mbar to 0 mbar (or hPa or cmH_2O)
Occlusion pressure P0.1	Range 0 to -25 mbar (or hPa or cmH_2O)
Waveform displays	Airway pressure Paw (t) -30 to 100 mbar (or hPa or cmH_2O) Flow (t) -180 to 180 L/min Volume V (t) 2 to 3,000 mL CO_2 (t) 0 to 120 mmHg

Technical Data

Alarms / Monitoring

Expiratory minute volume (MVe)	High / Low
Airway pressure (Paw)	High
Inspiratory O_2 concentration (FiO_2)	High / Low
End-tidal CO_2 concentration ($etCO_2$)	High / Low
Respiratory rate (RR)	High
Volume monitoring (VT)	High / Low
Apnoea alarm time (Tapn)	5 to 60 seconds, Off
Disconnection alarm time (Tdiscon)	0 to 60 seconds

Performance data

Control principle	Time-cycled, volume-constant, pressure-controlled
Length of intermittent PEEP	1 to 20 expiratory cycles
Medication nebulization	For 5, 10, 15, 30 minutes, continuously (∞)
Inspiratory flow	Max. 180 L/min, BTPS
Base flow, adults	2 L/min
Base flow, pediatric patients	3 L/min
Base flow, neonates	6 L/min
Inspiratory valve	Opens if medical compressed air supply fails (supply gas flow is not sufficient to provide the inspiratory flow required), enables spontaneous breathing with ambient air.

Endotracheal suction

Disconnection detection	Automatic
Reconnection detection	Automatic
Preoxygenation	Max. 3 minutes
Active suction phase	Max. 2 minutes
Postoxygenation	Max. 2 minutes
Factor for paediatric patients and neonates	1 to 2
Supply system for spontaneous breathing and Psupp	Adaptive CPAP system with high initial flow

Operating data

Mains power supply

Electric power inlet	100 V to 240 V, 50/60 Hz

Current consumption

At 230 V	Max. 1.3 A
At 100 V	Max. 3.0 A
Inrush current	Approx. 8 to 24 A peak Approx. 6 to 17 A quasi RMS

Power consumption

Maximum	300 W
During ventilation, without charging the battery	Approx. 100 W ventilation unit with display unit Approx. 180 W with GS500

Gas supply

O_2 positive operating pressure	2.7 to 6.0 bar (or 270 to 600 kPa or 39 to 87 psi)
Air operating pressure	2.7 to 6.0 bar (or 270 to 600 kPa or 39 to 87 psi)

Battery details

Internal battery of ventilation unit (without PS500)	Type NiMH battery, sealed Exchange interval 2 years

Technical Data

Battery runtime	Without GS500 30 minutes With GS500 15 minutes
Batteries in the PS500 power supply unit	Type LFP batteries Exchange interval 4 years
Battery runtime	Without GS500 240 minutes With GS500 120 minutes
Automatic switch over from internal to external battery	
Battery test available	
The battery runtime applies when the batteries are fully charged and new and ventilation is typical.	

Screen values	
Evita V600 diagonal screen size	15.6 inches
Input / Output ports	– 3 external RS232 (9-pin) connectors – 4 USB ports for data collection – 1 LAN port – 1 HDMI port
Touchscreen technology	Capacitive touchscreen with glass front
Aspect ratio	16:9
Resolution	1366 x 768 pixels
Digital machine output	Digital output and input via an RS232 C interface Dräger MEDIBUS® and MEDIBUS®X

1 BIPAP, trademark used under license. ATC®, trademarked by Dräger. AutoFlow®, trademarked by Dräger.
BTPS – Body Temperature Pressure Saturated. Measured values relating to the conditions of the patient lung 37° C (98.6° F), steam-saturated gas, ambient pressure.
1 mbar = 100 Pa

Some functionalities are available as an option.

CORPORATE HEADQUARTERS
Drägerwerk AG & Co. KGaA
Moislinger Allee 53–55
23558 Lübeck, Germany
www.draeger.com

Manufacturer:
Drägerwerk AG & Co. KGaA
Moislinger Allee 53-55
23542 Lübeck, Germany

REGION EUROPE
Drägerwerk AG & Co. KGaA
Moislinger Allee 53–55
23558 Lübeck, Germany
Tel +49 451 882 0
Fax +49 451 882 2080
info@draeger.com

REGION MIDDLE EAST, AFRICA
Drägerwerk AG & Co. KGaA
Branch Office
P.O. Box 505108
Dubai, United Arab Emirates
Tel +971 4 4294 600
Fax +971 4 4294 699
contactuae@draeger.com

REGION ASIA PACIFIC
Draeger Singapore Pte. Ltd.
61 Science Park Road
The Galen #04-01
Singapore 117525
Tel: +65 6872 9288
Fax: +65 6259 0398
asia.pacific@draeger.com

REGION CENTRAL AND SOUTH AMERICA
Dräger Indústria e Comércio Ltda.
Al. Pucurui - 51 - Tamboré
06460-100 - Barueri - São Paulo
Tel. +55 (11) 4689-4900
relacionamento@draeger.com

Locate your Regional Sales Representative at:
www.draeger.com/contact